

ASHWIN RAJAN R

Cloud & Backend Developer | Python • Go • AWS • Docker • Kubernetes

ashwinrajanr@gmail.com | +91-7010704499 | [linkedin.com/in/ashwin-rajan-rajamanickam](https://www.linkedin.com/in/ashwin-rajan-rajamanickam) | github.com/ashwinrajan-159

TECHNICAL SKILLS

Languages: Python, Go, Java, SQL, Bash

Cloud & DevOps: AWS (EC2, S3), Docker, Docker Compose, Kubernetes, GitHub Actions (CI/CD), Nginx, Linux

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, Celery, OpenAPI

Databases & Messaging: PostgreSQL, MySQL, Redis, Kafka, Alembic

Testing & Tools: pytest, Ruff, Bandit, Git

Integrations: Razorpay, Gmail API, Webhooks

EXPERIENCE

SusStudio — Backend Engineer Intern | Remote

Aug 2026 – Present

- Developing backend modules for **Autom8**, a multi-tenant AI automation SaaS built with FastAPI and PostgreSQL, supporting CRM, lead intelligence, and omni-channel communication workflows.
- Built the CRM lead-capture API (**4 REST endpoints**) that ingests inbound leads and attributes them to clients and campaigns within each tenant's isolated workspace.
- Integrated **Razorpay** for subscription payments and top-ups, and the **Gmail API** for client email.

ZettaOne — Software Engineer Intern | Bengaluru, India

Jun 2026 – Jul 2026

- Owned model selection for a document-processing feature by benchmarking **four OCR/vision systems** (Google Vision API, PaddleOCR, Granite, DreamMoon) across accuracy, latency, and cost, producing the comparison that determined the production model.

PROJECTS

Flimo — Cloud Search Service & Deployment Pipeline (deployed)

2025

Python, FastAPI, FAISS, Docker, Nginx, AWS EC2, GitHub Actions | [GitHub](#)

- Built a FastAPI search API backed by a FAISS vector index, with ingestion pipelines pulling data from the TMDB and YouTube APIs.
- Deployed to AWS EC2 behind an Nginx reverse proxy using a multi-stage Docker build; a GitHub Actions pipeline (push → SSH → build → health check) automates releases and cut deployments to **68 seconds**.
- Served HTTPS through a Cloudflare Tunnel run as an auto-restarting systemd service.

TrustLens — Loan Fraud Detection Platform | Canara Bank SuRaksha Hackathon 2.0

2026

Python, FastAPI, PostgreSQL, Celery, Redis, Kafka, React, Docker | [GitHub](#)

- Built a containerised platform for automated loan-fraud assessment: an async FastAPI REST API and React frontend over a 10-stage pipeline from OCR to risk scoring, run with Docker Compose.
- Moved long-running processing into Celery workers; made each stage idempotent so failed jobs retry without duplicate writes, and used a transactional outbox so database changes and Kafka events commit together.
- Enforced security policy in code: JWT with refresh-token rotation, Argon2 hashing, TOTP MFA, four-role RBAC, and startup checks that block production runs with weak secrets; CI runs pytest, Ruff, Bandit, and secret scanning.

CrowdWise — Traffic Deployment Planning Service (deployed)

2026

Python, FastAPI, LightGBM, Docker, pytest | [GitHub](#) • [Live Demo](#)

- Built and deployed a FastAPI service that turns a city event calendar into ranked congestion chokepoints and officer deployment plans, with REST endpoints for prediction, retraining, CSV upload, and event ingestion.
- Deployed to Render from declarative config (render.yaml, Procfile, .env.example) with a health-check endpoint, secrets kept out of the repository, and a scheduled job that re-forecasts as new events arrive.
- Estimated a **23.3% reduction in vehicle-hours of delay** via historical replay of **8,173** Bengaluru Traffic Police events; 12-test pytest suite.

EDUCATION

Vellore Institute of Technology, Bhopal

Sep 2023 – May 2027

B.Tech, Computer Science and Engineering | **CGPA: 8.36/10**

Relevant Coursework: Cloud Computing, Distributed Systems, Operating Systems, Computer Networks, DevOps

ACHIEVEMENTS & CERTIFICATIONS

- **AWS Certified Cloud Practitioner**, 2026.
- **Canara Bank SuRaksha Cyber Hackathon 2.0 (2026)**: shortlisted for the prototype phase with TrustLens; **130 of 1,560 teams** selected.
- **Top 100**, Zelestra × AWS ML Ascend Challenge.